FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: acdcwave
Model status: EXPLICIT_DOCUMENTED_AC_DC_WAVEFORM_DUTY
Captured UTC: 2026-10-26T06:04:39.147Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: frequencyHz | Entered value: 60
Input path: durationS | Entered value: 0.1
Input path: inceptionAngleDeg | Entered value: 40
Input path: initialCurrentCondition | Entered value: ZERO_PRIOR_CURRENT
Input path: sources[0].id | Entered value: UTILITY
Input path: sources[0].sourceReference | Entered value: ILLUSTRATIVE ONLY: not utility/OEM trace
Input path: sources[0].dcTimeConstantReference | Entered value: ILLUSTRATIVE ONLY: not measured equivalent X/R
Input path: sources[0].dcTimeConstantS | Entered value: 0.025
Input path: sources[0].samples[0].timeS | Entered value: 0
Input path: sources[0].samples[0].symmetricalRmsA | Entered value: 14000
Input path: sources[0].samples[0].phaseDeg | Entered value: -30
Input path: sources[0].samples[1].timeS | Entered value: 0.05
Input path: sources[0].samples[1].symmetricalRmsA | Entered value: 10000
Input path: sources[0].samples[1].phaseDeg | Entered value: -30
Input path: sources[0].samples[2].timeS | Entered value: 0.1
Input path: sources[0].samples[2].symmetricalRmsA | Entered value: 7000
Input path: sources[0].samples[2].phaseDeg | Entered value: -30
Input path: sources[1].id | Entered value: GENERATOR
Input path: sources[1].sourceReference | Entered value: ILLUSTRATIVE ONLY: not generator decrement or fault model
Input path: sources[1].dcTimeConstantReference | Entered value: ILLUSTRATIVE ONLY: not OEM rotor/DC time constant
Input path: sources[1].dcTimeConstantS | Entered value: 0.08
Input path: sources[1].samples[0].timeS | Entered value: 0
Input path: sources[1].samples[0].symmetricalRmsA | Entered value: 4000
Input path: sources[1].samples[0].phaseDeg | Entered value: 20
Input path: sources[1].samples[1].timeS | Entered value: 0.1
Input path: sources[1].samples[1].symmetricalRmsA | Entered value: 1500
Input path: sources[1].samples[1].phaseDeg | Entered value: 20

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: EXPLICIT_DOCUMENTED_AC_DC_WAVEFORM_DUTY
Calculated output path: model | Value: Piecewise-linear complex RMS phasor traces and source-specific exponential decaying DC offset, with documented common fault inception phase and zero pre-fault current
Calculated output path: frequencyHz | Value: 60
Calculated output path: durationS | Value: 0.1
Calculated output path: inceptionAngleDeg | Value: 40
Calculated output path: initialCurrentCondition | Value: ZERO_PRIOR_CURRENT
Calculated output path: sourceCount | Value: 2
Calculated output path: sourceProvenance[0].id | Value: UTILITY
Calculated output path: sourceProvenance[0].sourceReference | Value: ILLUSTRATIVE ONLY: not utility/OEM trace

Calculated output path: sourceProvenance[0].dcTimeConstantReference | Value: ILLUSTRATIVE ONLY: not measured equivalent X/R
Calculated output path: sourceProvenance[0].dcTimeConstantS | Value: 0.025
Calculated output path: sourceProvenance[0].initialDCOffsetA | Value: -19498.19872896053
Calculated output path: sourceProvenance[1].id | Value: GENERATOR
Calculated output path: sourceProvenance[1].sourceReference | Value: ILLUSTRATIVE ONLY: not generator decrement or fault model
Calculated output path: sourceProvenance[1].dcTimeConstantReference | Value: ILLUSTRATIVE ONLY: not OEM rotor/DC time constant
Calculated output path: sourceProvenance[1].dcTimeConstantS | Value: 0.08
Calculated output path: sourceProvenance[1].initialDCOffsetA | Value: -2828.42712474619
Calculated output path: totalInitialDCOffsetA | Value: -22326.625853706722
Calculated output path: waveformI2tA2S | Value: 23143646.66048021
Calculated output path: equivalentAsymmetricalRmsA | Value: 15213.0360745251
Calculated output path: symmetricalRmsPhasorI2tA2S | Value: 15536194.869099773
Calculated output path: absolutePeakInstantaneousA | Value: 39985.80203786386
Calculated output path: timeAtAbsolutePeakS | Value: 0.007153674229325416
Calculated output path: quadratureSubintervals | Value: 48
Calculated output path: quadratureRelativeDisagreement | Value: 1.9754802668518058e-10
Calculated output path: peakRefinements | Value: 592
Calculated output path: atStartInstantaneousA | Value: 0
Calculated output path: atClearingInstantaneousA | Value: 9642.27963055042
Calculated output path: approved | Value: false
Calculated output path: limitations[0] | Value: Not an IEC 60909-0 equivalent thermal factor or certified making/breaking current: no standard voltage/impedance source correction, breaker contact timing, arc restrike, CT saturation, harmonics, waveform discontinuities or inverter controls
Calculated output path: limitations[1] | Value: Every symmetrical source phasor history and DC time constant must be independently documented; no generator/motor/IBR physics is inferred
Calculated output path: limitations[2] | Value: Assumes zero pre-fault current and independent source DC decay from one common fault inception angle; cannot be applied to loaded circuits or arbitrary current-source inverter dynamics without validating those assumptions
Calculated output path: limitations[3] | Value: Output is a prospective model screening waveform; verify model and protection device suitability before engineering use

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: acdcwave
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:39.147Z
Evidence / limitation: outputs.limitations[0] | Value: Not an IEC 60909-0 equivalent thermal factor or certified making/breaking current: no standard voltage/impedance source correction, breaker contact timing, arc restrike, CT saturation, harmonics, waveform discontinuities or inverter controls
Evidence / limitation: outputs.limitations[1] | Value: Every symmetrical source phasor history and DC time constant must be independently documented; no generator/motor/IBR physics is inferred
Evidence / limitation: outputs.limitations[2] | Value: Assumes zero pre-fault current and independent source DC decay from one common fault inception angle; cannot be applied to loaded circuits or arbitrary current-source inverter dynamics without validating those assumptions
Evidence / limitation: outputs.limitations[3] | Value: Output is a prospective model screening waveform; verify model and protection device suitability before engineering use